

Ageing and PLHIV

- The age of the population of PLHIV in Jamaica is steadily increasing
 - increased life expectancy due to the availability of ART
 - increasing contribution of the older age group to the number of new infections occurring.

- People aging with HIV share many of the same health concerns as the general population aged 50 and older:
 - multiple chronic diseases or conditions
 - the use of multiple medications
 - changes in physical and cognitive abilities
 - increased vulnerability to stressors

- While effective HIV treatment has decreased the likelihood of AIDS-defining illnesses among people aging with HIV, many HIV-associated non-AIDS conditions occur
 - Cardiovascular disease
 - Diabetes
 - Renal disease
 - Cancer

■ **Poor CD4 count recovery with HAART in the elderly population**

- the natural ageing process associated with decline in immune system seen in aging coupled with the immune depletion of HIV.
- the elderly population generally have fibrotic lymph nodes with poor regenerative capacity.

This effect may not be persistent and CD4 count responses may improve over time with ART

- **Later presentation to treatment and care with corresponding low CD4 counts seen**
 - elderly patients often don't consider themselves at risk for HIV
 - safe sexual practices must be emphasized.

- **High frequency of comorbid illness with high potential for treatment related toxicities.**

- Gradual decline in the function of the liver and the kidneys, decreasing the metabolism of pharmacologic agents resulting in high drug levels and increased toxicity.

- **Increased risk of drug-drug interactions with polypharmacy** required for comorbid illnesses.
- **Misinterpretation of some non-specific symptomatology of HIV** as due to aging, e.g. fatigue.

■ **Cancer risk**

- No studies have proven the need for more stringent cancer screening protocols for non-AIDS defining cancers in the PLHIV population.
- However, elderly PLHIV are at increased risk for some non-AIDS defining cancers, e.g. primary lung or liver cancers.

■ **Bone disease**

- osteopenia and osteoporosis.
- Vitamin D and Calcium supplementation should be considered in the management of the elderly PLHIV

- **Neurological and psychosocial health**
 - age and HIV infection are independent risk factors for the development of dementia.
 - High viral load and low CD4 counts have also been associated with an increased risk.

Interplay of Age With Morbidity

- Risk of “comorbidities” increases as individuals get older
- HIV does not cause these illnesses
- However, HIV increases or accelerates the risk

Aging and Non-HIV Comorbidities

- Non-HIV comorbidities become increasingly important in HIV care, particularly
 - Diabetes mellitus
 - Cardiovascular disease
 - Non-AIDS malignancies
 - Renal disease/hypertension
 - Osteoporosis

- Majority of the risk from “standard” risk factors (eg, **genetics, environment, diet**)
 - Increasing age and substance use (including alcohol and tobacco) add to risk
- HIV adds to risk
 - Is associated with CD4+ cell count, low CD4+ count, or higher viral loads
 - Role of chronic viral infection
 - Role of chronic inflammation

HIV and Diabetes Mellitus

- Patients may present to care already being diabetic or it may develop after commencing ART.
- Several ART agents have been associated with hyperglycemia and diabetes.
 - Offending agents include: - PI: most commonly associated with Lopinavir

- Glucose assessments should be conducted as part of the initial investigation of all PLHIV presenting to care.
- Evaluations should also be repeated 3-6 months after commencing or switching ART.
- HbA1c levels can also be utilized, but may underestimate hyperglycemia in PLHIV

How do you avoid Type 2 DM?

- Screen
- Keep at NORMAL body weight
- Exercise regularly
- Reduce starch content in diet

Best practices for T2DM in HIV

- Adhere to national guidelines and goals, encourage diet and exercise
 - FBG 4.5 – 6.1mmol/L, 2hrpp 5.0-7.8mmol/L
 - LDL <2.6mmol
 - BP 120/80
 - No proteinuria
 - Normal BMI, normal waist
- Caution with setting targets for A1C

Kim et al. Monitoring hiV-infected

Patients with diabetes: hemoglobin a1c, Fructosamine, or glucose? *Clinical Medicine Insights: Endocrinology and Diabetes*

Treatment for T2DM in HIV

- Due to the underlying pathology of insulin resistance in this population, insulin sensitizers such as metformin (treatment of choice in overweight individuals), or thiazolidinedione (pioglitazone) can be utilized.

Interactions ARVs with antidiabetic Meds

ARV class	Sulphonylureas, DPP4 inhibitors, thiazolidines, metformin, insulin	
NRTI	No interactions	
NNRTI	Possibly EFV with glyburide	
PI	RTV – CYP3A4 inhibitor that increases the levels of all classes(?how much) except insulin	
INSTI	DTG – increases levels of metformin by 66%	

HIV and Cardiovascular disease

- Many studies have supported the increased risk of cardiovascular disease in PLHIV.
 - factors associated with HIV itself , ART and traditional risk factors all play a role
 - control of traditional, modifiable risk factors continues to have the greatest impact on cardiovascular disease (diet, exercise, tobacco smoking)

- Studies related to Zidovudine and dilated cardiomyopathy have been published, although the risk remains low.
- Studies have been conducted to investigate the association between Abacavir and myocardial infarction and the latest meta-analysis failed to show a conclusive result.

RE: HIV infection and CVD risk

- HIV infection is an independent risk factor for CVD
- ART partially reverses the risk associated with HIV infection
- LPV/r causes dyslipidemia – choose other PI in the face of established CVD
- Atazanavir and Darunavir have the lowest impact on lipid parameters.
- HIV-infected patients should be managed with aggressive primary prevention
 - Control BP, lipids, smoking cessation

CAUTION: Drug –drug interactions PIs with statins and CCBs

ARV Class	Statin, CCB, aspirin/clopidogrel
NRTI	No interaction ABC – not recommended for now
NNRTI	No interaction
PI	Statins – use only 10mg Atorvatsatin or 5 mg Rosuvatstain Or Pravastatin. Simvastatin absolutely C/I
INSTI	No interactions

HIV and Kidney Disorders in the ART Era

- Increasing prevalence
 - Growing burden of CKD related to comorbid diabetes and hypertension
- Independent predictor of cardiovascular events and mortality
- Implications with respect to drug dosing and toxicity
 - Includes antiretroviral drugs

Survey of Chronic Renal Failure in Jamaica

N= 605

	<i>Percent prevalence with chronic renal failure</i>
Hypertension	60.8
Diabetes Mellitus	31.4
Primary chronic glomerulonephritis	6.0
Obstructive nephropathy	3.8
Adult polycystic kidney disease	3.5
Systemic lupus erythematosus	2.3
Nephrotoxic agents	2.3
Genitourinary neoplasm	2.0
HIV/AIDS	1.8
Genitourinary chronic pyelonephritis	0.8
Interstitial nephritis	0.7
<u>Sickle cell disease</u>	<u>0.7</u>

Summary: Screening and Management of Kidney Disease in HIV-Positive Patients

- **Screen for Renal disease annually with Serum Creatinine (eGFR) and Urine dipstick for proteinuria**
- **Appropriate treatment selection and/or dose reduction** is warranted for ART (mainly NRTI) or other medications that are primarily eliminated by the kidneys
- **Referral to a nephrologist** is recommended for patients with:
 - Proteinuria (grade 1+ by dipstick analysis)
 - Reduced renal function ($\text{GFR} < 60 \text{ mL/min/1.73 m}^2$)
 - Need for treatment of CKD



National Kidney Foundation™

Press Here for Clinical Use and Formula

S_{Cr}:

umol/L

Age:

years

Gender:

Male

Female

Race:

Black

Other

eGFR:

Clear All

mL/min/1.73 m²



Calculators



About GFR



About CKD

Osteoporosis in HIV-Positive Patients

- Osteoporosis and fractures are common in HIV-positive patients and will increase with aging
- Risk factors include
 - HIV: chronic infection, ART (TDF, certain PIs)
 - Traditional: smoking, alcohol, low weight
- Screening: dual-energy x-ray absorptiometry should be considered in all HIV-positive postmenopausal women and in men aged older than 50 yrs
- African Heritage have ↑BMD, but still at risk....
- Treat according to National Guidelines.

Cancer

- Unlikely that HIV itself or its treatment contributes to increasing prevalence of Cancer in HIV+ patients

Screen for common cancers in JA:

- Men- PSA, Stool FOB/ colonoscopy at age 50
- Women – Breast, Cervical, Colon

Factors Associated With Late Diagnosis in Older Patients

- Compared with younger patients, older patients:
 - Are more likely to present late for HIV diagnosis and care^[1-3]
 - Are more likely to have been diagnosed with HIV infection while presenting with other illnesses^[1]
 - Have poorer routine access to HIV testing
- Physicians less likely to discuss HIV/AIDS and related risk factors with older patients
- Patients with undocumented HIV infections significantly more likely to be older than 55 yrs of age

Starting ART in a Newly Diagnosed Older Patient

- Assess readiness for ART
 - Disease characteristics
 - Psychological readiness
 - Adjustment to diagnosis
 - Ability to disclose
- Evaluate comorbidity and other concurrent drugs to guide choice of ART regimen
- Sexual History!!

SUMMARY

- ART has improved the longevity of PLHIV
- The population of PLHIV on ART is aging, leading to increasing numbers of older patients with comorbidities; T2DM, HTN, CVD, CKD, Bone
- HIV impacts the onset of comorbidities, but traditional risk factors still very important
- HIV is an independent risk factor for CVD + CKD
- Drug-drug interactions become more common with increasing comorbidities
- Keeping the VL 'undetectable' and CD4>500 protects the patient against the risk conferred by HIV



Interactions with Lenacapavir (Sunlenca®) now available - [click here](#) for more details

Looking for interactions with COVID-19 therapies, including Paxlovid? [Click here for covid19-druginteractions.org](#)

If a drug is not listed below it cannot automatically be assumed it is safe to coadminister

HIV Drugs	Co-medications	Drug Interactions
Search HIV drugs...	Search co-medications...	☐ Check HIV/ HIV drug interactions
☒ A-Z ☐ Class ☐ Trade	☒ A-Z ☐ Class ☐ Trade	Drug Interactions will be displayed here
Selected HIV Drugs will be displayed here.	Selected Co-medications will be displayed here.	

- 70 yo lady on TLD x 3yrs
- CD4 = 380(28%) VL=150 cp/ml
- No Sig PMH; No complaints
- BP 140/90, wt 46kg
- Creat 138 μ mol/ml

- $\text{eGFR} = 40\text{ml/min}/1.73\text{m}^2$

- STOP TDF, and never use again
- Cont DTG
- Add ABC 600mg od (or AZT 300mg bd)
- Adjust 3TC for Cr Cl; >50: 300mg od
 - CrCl 30-49: 150mg od
 - CrCl 15-29: 100mg od
 - CrCl 5 -14: 50mg od
 - CrCl <5 : 25mg od

Thank
you